

Joseph Dobbelaar

Software Engineer & Computer Science Alum From WPI

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EDUCATION

Worcester Polytechnic Institute — *Bachelors in Computer Science*

2020–2024

Studied Computer Science at WPI with a focus on collaborative software development and thoughtful user interface design, as well as taking classes related to artificial intelligence and machine learning. Completed projects such as: training machine learning models, building websites and mobile applications, designing software for use in a hospital, virtual environments, and more. Worked in the WPI Data Visualization lab. President of the Guitar Club.

SKILLS

- ★ Confident Full-Stack Software Engineer
- ★ Agile Scrum
- ★ React.js, React Native, Express.js, Node.js, .NET
- ★ JavaScript, TypeScript, Java, C#, C, C++, Python
- ★ Machine Learning & Data Analysis
- ★ UI / UX Methodology, Tools, & Libraries
- ★ Object Oriented Design Patterns
- ★ Code Documentation & Diagrams
- ★ SQL and noSQL Databases
- ★ AWS, Firebase, Linux Console, Kubernetes
- ★ Test-Driven Software Development
- ★ Data Visualization Tools & Techniques

EXPERIENCE

WPI Data Visualization Lab — *Software Engineer* — (February 2024 – Current)

Working on the reVISit platform, a web-based toolkit for creating and conducting data visualization experiments.

Freelance Web Development — *Self Employed* — (November 2023 – Current)

Building websites for individuals and businesses using “Web-Legos,” a set of full-stack JavaScript libraries that I’ve developed to ensure security, accessibility, and rapid iteration. Clients can edit site content on their own when signed in with Google.

UMass Chan Medical School — *Software Engineer* — (April 2023 – September 2023)

Single-handedly built the HIPAA compliant computing infrastructure for UMass Chan’s PUPSys: a system for preventing pressure ulcers in hospital patients.

Citrus Financial — *Chief Technology Officer* — (June 2022 – April 2023)

Started a company around a mobile app aimed at simplifying how friends, couples, and roommates split shared expenses. Built the app alone in React Native with Google Firebase and Express.js as the back-end. Spoke with possible investors but, ultimately, my partners graduated and their priorities shifted.

Sentaca, an IBM Company — *Full-Time Software Engineer Intern* — (June 2022 – August 2022)

Created virtual machines and networks using AWS CloudFormation. Learned to build Kubernetes clusters. Laid foundation for the migration of a monolithic software to a cloud-based microservices architecture. Hands-on experience with AWS cloud computing technologies and design patterns.

Blueport Commerce — *Full-Time Product Reliability Intern* — (June 2021 – August 2021)

Wrote and repaired automated tests in C#, ensuring that all code in production was bug-free. Gained some SQL experience. Became familiar with Agile Scrum methodology, code documentation best practices, and other aspects of team development.

Empow Studios — *Lead Instructor & Healthcare Supervisor* — (April 2018 – May 2022)

Worked as a Lead Instructor and Healthcare Supervisor for children grades 2–8 teaching Java, Python, game design, animation, 3D printing, and more. Was the only instructor for the “Python Advanced Team,” a class on program design concepts like team planning, git usage, and code documentation.

Zaniac — *Instructor* — (April 2017 – February 2018)

Taught STEM through 3D printing, coding, math tutoring, and more to children K–8, until their closing in February 2018

PROJECTS

Executive Functioning Coaching Suite — 2024

Built a web app that integrates all the tools necessary for executive functioning coaching. Coaches can manage student assignments, send invoices, assign forms, and more. Students can track assignments in a list or Gantt chart, pay invoices, access shared documents, and view assigned tools and strategies. Parents can link their accounts to monitor progress and pay invoices. I’m exploring options to transition it to a software-as-a-service model.

Data Visualization Literacy App — 2023 & 2024

Worked with a team to build a mobile app aimed at assessing and teaching data visualization literacy.

Web-Legos — 2023 & 2024

Developing my own set of full-stack JavaScript libraries for quick, accessible, and secure web development. I’ve used these tools to build websites for clients, as well as my personal portfolio.

WPI Student Success Handbook — 2023

Built a mobile-friendly website for incoming freshmen at WPI to access their resources, designed to ensure that all students, including those from diverse backgrounds, can access and benefit from campus resources.

Árboles Mágicos — 2023

Collaborated with a team to build an app in .NET MAUI that allows users to mark the locations of flowering trees on a public map.

Medical Tracker — 2022

Worked on a team using Agile Scrum methodology to create an application for hospital employees to create and organize tasks.

WICN Virtual Environment — 2021

Worked on a team in collaboration with Worcester radio station to create a virtual space for interacting socially during live music performances attended online.